The supersingular factorial constant: a proof of $E_{p+1}^{(p-1)/2} \equiv \left(\frac{6}{p}\right)\left(\frac{p-1}{2}\right)!\,\Delta^{(p^2-1)/24} \pmod{p, E_{p-1}}$

Research note on Conjecture 1.5 of D. Jordan,
Supersingular poles and signed residues of prime-level Fricke Hauptmoduln, v1.4

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Abstract

For a prime $p \geq 5$ put $h = (p-1)/2$, $k = (p^2-1)/24$, $A = E_{p-1} \bmod p$ and $B = E_{p+1} \bmod p$. We prove that $B^h - \left(\frac{6}{p}\right) h!\,\Delta^k$ is divisible by A in $\mathbb{F}_p[E_4, E_6]$; equivalently, $c(E) = B(E,\omega)^h/\Delta(E,\omega)^k$ equals $\left(\frac{6}{p}\right) h!$ for every supersingular elliptic curve $E/\overline{\mathbb{F}}_p$. This is Conjecture 1.5 of the manuscript. The proof is uniform in p and does not use constancy of c, Baker's congruence, isogeny graphs, the eta–Frobenius theorem or CM points. Its main ingredient is the weight-$\frac{p+1}{2}$ form $F = \eta^{p+1}$ on $\Gamma(12)$. Modulo p one has $F \equiv \eta(\tau)\eta(p\tau)$, so its quadratic twist $\theta^{(p-1)/2}F$ is an explicit combination of $A^{(p+1)/2}F$ and F^p. Iterating $\theta f = A\,\partial f + \frac{w}{12}Bf$ shows that $\theta^{(p-1)/2}$ multiplies values at supersingular points by $-(B/12)^h/h!$. Comparing the two expressions gives the theorem, because $F^{p-1} = \Delta^k$. Constancy of c and Baker's congruence follow as corollaries. So do the eta-character identity of the manuscript and a signed refinement of the Kaneko–Zagier/de Shalit formula for $ss'_p(j)$. The proof is checked by exact computation for all 166 primes $5 \leq p \leq 997$ along four independent routes.

1 Statement

Throughout, $p \geq 5$ is prime, $h = (p-1)/2$, $k = (p^2-1)/24$ and $m = (p+1)/2$. The Eisenstein series $E_w = 1 - \frac{2w}{B_w}\sum_{n\geq 1}\sigma_{w-1}(n)q^n$ have constant term 1, $\Delta = q\prod_{n\geq 1}(1-q^n)^{24} = (E_4^3 - E_6^2)/1728$, and

$$A = E_{p-1} \bmod p, \qquad B = E_{p+1} \bmod p.$$

For an elliptic curve E over a field of characteristic p with nowhere-vanishing invariant differential ω, $f(E,\omega)$ denotes the value of a (Katz) modular form f; for weight w one has $f(E,u\omega) = u^{-w}f(E,\omega)$. Since B^h and Δ^k both have weight $(p^2-1)/2$, the ratio

$$c(E) = \frac{B(E,\omega)^h}{\Delta(E,\omega)^k}$$

is independent of ω. $\left(\frac{a}{p}\right)$ is the Legendre symbol, $\chi_{12}(n) = \left(\frac{12}{n}\right)$ the Kronecker symbol (so $\chi_{12}(n) = 1$ for $n \equiv \pm 1$, -1 for $n \equiv \pm 5 \pmod{12}$, 0 otherwise), and $\theta = q\frac{d}{dq} = \frac{1}{2\pi i}\frac{d}{d\tau}$.

Theorem 1.1. *Let $p \geq 5$ be prime.*

(a) *For every supersingular elliptic curve $E/\overline{\mathbb{F}}_p$ and every nonzero invariant differential ω,*

$$B(E,\omega)^{(p-1)/2} = \left(\frac{6}{p}\right)\left(\frac{p-1}{2}\right)!\,\Delta(E,\omega)^{(p^2-1)/24} \quad \text{in } \overline{\mathbb{F}}_p.$$

(b) *Equivalently,* $E_{p+1}^{(p-1)/2} \equiv \left(\frac{6}{p}\right) \left(\frac{p-1}{2}\right)! \, \Delta^{(p^2-1)/24} \pmod{(p, E_{p-1})}$ *in* $\mathbb{Z}_{(p)}[E_4, E_6]$, *i.e. A divides* $B^h - \left(\frac{6}{p}\right) h! \Delta^k$ *in* $\mathbb{F}_p[E_4, E_6]$.

In particular Conjecture 1.5 of the manuscript holds: $c_p = \left(\frac{6}{p}\right) h! \bmod p$.

The equivalence of (a) and (b), including the behaviour at $j = 0$ and $j = 1728$, is Lemma 2.2. The proof of (a) occupies Section 3. It is self-contained apart from the standard inputs listed in Section 2.

Corollary 1.2. *Let* $p \geq 5$.

(i) (Constancy; manuscript Corollary 1.4.) $c(E)$ *is the same element* $c_p = \left(\frac{6}{p}\right) h!$ *of* $\mathbb{F}_p$ *for every supersingular* E.

(ii) (Baker's congruence [1, Theorem 1.4].) $B^{p-1} \equiv -\left(\frac{-1}{p}\right) \Delta^{(p^2-1)/12} \pmod{A}$.

(iii) (Eta character; manuscript Theorem 1.3.) *For every prime* $\ell \neq p$ *and every supersingular point* x *of* $X_0(\ell)_{\overline{\mathbb{F}}_p}$, $\left(\Delta(\tau)/\Delta(\ell\tau)\right)^k(x) = \left(\frac{\ell}{p}\right)$.

Proof. (i) is Theorem 1.1(a). For (ii), Wilson's theorem gives $h!^2 \equiv (-1)^{h+1} = -\left(\frac{-1}{p}\right)$, so squaring (a) gives $B^{p-1} = -\left(\frac{-1}{p}\right) \Delta^{2k}$ at every supersingular point, and Lemma 2.2 applies. For (iii), let $x = (E, C)$ with $\pi : E \to E' = E/C$ and $\omega = \pi^* \omega'$. Robert's isogeny identity in the manuscript's normalization [8, Théorème B], [1, Lemma 2.2], manuscript Lemma 4.3, states $B(E', \omega') = \ell B(E, \omega)$ and $\Delta(\ell\tau)(x) = \ell^{-12} \Delta(E', \omega')$. It gives $c(E')/c(E) = \left(\frac{\ell}{p}\right) \left(\Delta/\Delta(\ell\tau)\right)^k(x)$, using $\ell^{12k} = 1$ and $\ell^h = \left(\frac{\ell}{p}\right)$; this is the computation in the manuscript's proof of Corollary 1.4. By (i) the left side is 1. □

Thus Corollary 1.2(iii) reverses the logical order of the manuscript. There, Theorem 1.3 and the connectivity of the isogeny graph give constancy. Here, constancy and Robert's identity give Theorem 1.3 for every ℓ. Explicit consequences on the j-line, on Weierstrass models and on the Legendre family are in Section 4.

2 Standard inputs

We use Katz's theory of modular forms over $\mathbb{Z}[1/N]$-algebras for the full level structure $\Gamma(N)$, $N \geq 3$, $p \nmid N$ [3, §§1.2–1.6], [2]. Let $X(N)^\zeta$ be the component of the compactified moduli scheme of triples (E, α, ω) on which the Weil pairing of the level structure is a fixed primitive Nth root of unity ζ. It is smooth and proper with geometrically connected fibres over $\mathbb{Z}[1/N, \zeta_N]$, and its complex points are $\Gamma(N)\backslash\mathfrak{H}^*$. Its standard cusp has q-expansion parameter $q^{1/N}$. We take $N = 12$ and work over $R = \mathbb{Z}_{(p)}[\zeta_{12}] \subset W(\overline{\mathbb{F}}_p)$.

Input S1 (q-expansion principle)**.** [3, §1.6]. A classical modular form of weight w on $\Gamma(N)$ whose Fourier expansion at ∞ has coefficients in R is a section of $\omega^{\otimes w}$ on $X(N)^\zeta_R$. If all its coefficients lie in pR, its reduction vanishes identically on the special fibre. In particular its value at every test object (E, α, ω) over $\overline{\mathbb{F}}_p$ is 0. Values commute with reduction modulo p.

Input S2 (Hasse invariant)**.** [3, §2.1], [9]. For $E/\overline{\mathbb{F}}_p$, $A(E, \omega) = E_{p-1}(E, \omega) \bmod p$ is the Hasse invariant of (E, ω). It vanishes if and only if E is supersingular. Its zeros are simple, and A is squarefree in $\mathbb{F}_p[E_4, E_6]$, with irreducible factors E_4 if $j = 0$ is supersingular, E_6 if $j = 1728$ is supersingular, and the forms $E_4^3 - j_0 \Delta$ (over $\overline{\mathbb{F}}_p$) for the supersingular $j_0 \neq 0, 1728$ [1, Theorem 1.2], [4, §3 and §10(a)].

Input S3 (Kummer congruences)**.** $E_{p-1} \equiv 1$ and $E_{p+1} \equiv E_2 = 1 - 24\sum \sigma_1(n)q^n$ coefficientwise modulo p. Both have p-integral coefficients by von Staudt–Clausen and Kummer.

Input S4 (Serre derivative)**.** If f is a modular form of weight w on a finite-index subgroup $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$, then $\partial_w f := \theta f - \frac{w}{12} E_2 f$ is a modular form of weight $w+2$ on Γ. If the Fourier coefficients of f lie in R then so do those of $\partial_w f$, since $p \geq 5$. On $\mathbb{Z}[\frac{1}{6}][E_4, E_6]$, ∂ is the derivation with $\partial E_4 = -E_6/3$ and $\partial E_6 = -E_4^2/2$, and $\partial\Delta = 0$ (Ramanujan).

Input S5 (Euler)**.** $\eta(\tau) = q^{1/24}\prod_{n\geq 1}(1-q^n) = \sum_{n\geq 1}\chi_{12}(n)\, q^{n^2/24}$ and $\eta^{24} = \Delta$.

Lemma 2.1. *η^2 is a holomorphic modular form of weight 1 on $\Gamma(12)$ with trivial character. Hence, for every even integer $r \geq 2$, $\eta^r \in M_{r/2}(\Gamma(12))$ with integral Fourier coefficients in $q^{1/12}$.*

Proof. η^2 is holomorphic and nonvanishing on $\mathfrak{H}$ and has positive order at every cusp. Write $\eta(\gamma\tau) = \nu(\gamma)(c\tau+d)^{1/2}\eta(\tau)$. Then $\eta^2(\gamma\tau) = \nu(\gamma)^2(c\tau+d)\eta^2(\tau)$, and ν^2 is a character of $\mathrm{SL}_2(\mathbb{Z})$. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(12)$. If $c = 0$ then $\gamma = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$ with $12 \mid b$, and $\eta^2(\tau+b) = e^{2\pi i b/12}\eta^2(\tau) = \eta^2(\tau)$. If $c > 0$, then c is even. The Petersson–Rademacher formula (manuscript eq. (4.1); [5, Ch. 4]) gives $\nu(\gamma)^2 = \left(\frac{c}{d}\right)^2 e\left(\frac{(a+d)c - bd(c^2-1) + 3d - 3 - 3cd}{12}\right)$. Here $\left(\frac{c}{d}\right)^2 = 1$, and the numerator is $\equiv 0 \pmod{12}$ because $12 \mid b$, $12 \mid c$ and $12 \mid d-1$. If $c < 0$, apply the formula to $-\gamma$, which has $a, d \equiv -1 \pmod{12}$. The numerator is then $\equiv 3(d'-1) \equiv -6 \pmod{12}$, so $\nu(-\gamma)^2 = -1$. Since $\eta^2((-\gamma)\tau) = \nu(-\gamma)^2(-c\tau - d)\eta^2(\tau)$, this gives $\nu(\gamma)^2 = +1$. The leading exponent of η^r is $r/24 \in \frac{1}{12}\mathbb{Z}$ for even r. □

The script `eta2_gamma12_check.py` confirms Lemma 2.1 numerically to 56 digits for random elements of $\Gamma(12)$ with $c > 0$ and $c < 0$.

Lemma 2.2 (divisibility criterion)**.** *Let $G \in \mathbb{F}_p[E_4, E_6]$ be isobaric. Then $A \mid G$ if and only if $G(E, \omega) = 0$ for every supersingular $E/\overline{\mathbb{F}}_p$ and every ω. Consequently (a)⇔(b) in Theorem 1.1. Moreover, with $\mathrm{ss}_p(j) = \prod_{E\ \mathrm{ss}}(j - j(E))$ and $R(j) = B^h/\Delta^k \in \mathbb{F}_p[j]$, (b) is equivalent to $R(j) \equiv \left(\frac{6}{p}\right) h! \pmod{\mathrm{ss}_p(j)}$.*

Proof. If $A \mid G$, then G vanishes at supersingular points by Input S2. Conversely, work over $\overline{\mathbb{F}}_p$ and use the factorization in Input S2. For supersingular $j_0 \neq 0, 1728$, the form $E_4^3 - j_0\Delta = (1 - \frac{j_0}{1728})E_4^3 + \frac{j_0}{1728}E_6^2$ is irreducible, since $\alpha X^3 + \beta Y^2$ with $\alpha\beta \neq 0$ is irreducible. Its zero set in $\mathbb{A}^2 \smallsetminus \{0\}$ is exactly the set of pairs $(E_4(E,\omega), E_6(E,\omega))$ with $j(E) = j_0$. If G vanishes there, this factor divides G. If $j = 0$ is supersingular, the zero set of the factor E_4 is $\{(0, y) : y \neq 0\}$, the set of values for curves with $j = 0$. Vanishing there means $G(0, E_6) = 0$, i.e. $E_4 \mid G$. The case $j = 1728$, with factor E_6, is the same. Since A is squarefree, $A \mid G$.

This settles the stack-multiplicity issue. On the j-line the zeros of A at $j = 0$ and $j = 1728$ have multiplicity $1/3$ and $1/2$. In the graded ring $\mathbb{F}_p[E_4, E_6]$ the corresponding factors E_4 and E_6 occur to the first power. The criterion is formulated in that ring, so vanishing at the points $(0, y)$, resp. $(x, 0)$, suffices.

For the last statement write $p - 1 = 12n + 4\delta + 6\varepsilon$ with $\delta, \varepsilon \in \{0, 1\}$. Then $A = E_4^\delta E_6^\varepsilon \Delta^n s(j)$ and $\mathrm{ss}_p = j^\delta(j - 1728)^\varepsilon s(j)$ [4, §3], [1, §6]. Every isobaric form of weight $12k$ is $\Delta^k r(j)$ with $\deg r \leq k$. Clearing the factors E_4^δ, E_6^ε with $E_4^3 = j\Delta$ and $E_6^2 = (j - 1728)\Delta$ shows that $A \mid \Delta^k r(j)$ if and only if $\mathrm{ss}_p \mid r$. □

3 Proof of Theorem 1.1

Put $F = \eta^{p+1}$. Since $p+1$ is even, Lemma 2.1 gives $F \in M_m(\Gamma(12))$, $m = \frac{p+1}{2}$, with coefficients in $\mathbb{Z}$. By Input S5,

$$F^{24} = \Delta^{p+1}, \qquad F^{p-1} = \eta^{p^2-1} = \Delta^k. \tag{3.1}$$

Lemma 3.1 (iterating θ)**.** *Define forms on* $\Gamma(12)$ *by* $G_0 = F$ *and*

$$G_{i+1} = E_{p-1}\,\partial_{w_i} G_i + \frac{w_i}{12}\,E_{p+1}\,G_i, \qquad w_i = m + i(p+1).$$

Then $G_i \in M_{w_i}(\Gamma(12))$ *has coefficients in* R*, and:*

(i) $G_i \equiv \theta^i F \pmod p$ *coefficientwise;*

(ii) $G_i = \tilde{c}_i\,E_{p+1}^i F + E_{p-1}R_i$ *with* $R_i \in M_{w_i-(p-1)}(\Gamma(12))$ *p-integral and* $\tilde{c}_i = \prod_{l=0}^{i-1}\frac{w_l}{12} \in \mathbb{Z}_{(p)}$*;*

(iii) $\tilde{c}_h \equiv 12^{-h}\prod_{l=0}^{h-1}(m+l) = 12^{-h}\,\frac{(p-1)!}{h!} \equiv -\frac{12^{-h}}{h!} \pmod p$.

Proof. Modularity and integrality follow by induction from Input S4. For (i), Input S3 gives $G_{i+1} \equiv \partial_{w_i} G_i + \frac{w_i}{12}E_2 G_i = \theta G_i$. For (ii), $G_{i+1} = E_{p-1}(\partial_{w_i}G_i + \frac{w_i}{12}E_{p+1}R_i) + \frac{w_i}{12}\tilde{c}_i E_{p+1}^{i+1}F$. For (iii), $w_l \equiv m + l \pmod p$, and $m + h - 1 = p - 1$, $m - 1 = h$. □

Lemma 3.2 (the twist of η^{p+1})**.** *As power series in* $q^{1/12}$ *over* $\mathbb{F}_p$*,*

$$\theta^h F \equiv \left(\frac{6}{p}\right)\left(F - \chi_{12}(p)\,F^p\right) \pmod p.$$

Consequently the form $D := G_h - \left(\frac{6}{p}\right)\left(E_{p-1}^m F - \chi_{12}(p)F^p\right) \in M_{pm}(\Gamma(12))$ *has all Fourier coefficients in* pR*.*

Proof. $\eta^p = q^{p/24}\prod(1-q^n)^p \equiv q^{p/24}\prod(1-q^{pn}) = \eta(p\tau)$, hence by Input S5

$$F \equiv \eta(\tau)\eta(p\tau) = \sum_{a,b\geq 1}\chi_{12}(a)\chi_{12}(b)\,q^{(a^2+pb^2)/24}.$$

On q^e, θ^h is multiplication by e^h. For $e = (a^2+pb^2)/24$ with $\chi_{12}(a) \neq 0$, one has $e \equiv a^2/24 \pmod p$ in $\mathbb{Z}_{(p)}$. So $e^h \equiv 0$ if $p \mid a$. If $p \nmid a$, then $e^h \equiv a^{p-1}24^{-h} \equiv \left(\frac{24}{p}\right) = \left(\frac{6}{p}\right)$. Therefore $\theta^h F \equiv \left(\frac{6}{p}\right)(F - \Sigma)$, where Σ is the sub-sum over $p \mid a$. Writing $a = pa'$ and using the multiplicativity of χ_{12},

$$\Sigma = \chi_{12}(p)\sum_{a',b}\chi_{12}(a')\chi_{12}(b)\,q^{p(b^2+pa'^2)/24} = \chi_{12}(p)\,\left(\eta(\tau)\eta(p\tau)\right)\big|_{\tau\mapsto p\tau} \equiv \chi_{12}(p)F^p,$$

because a series g over $\mathbb{F}_p$ satisfies $g(q^p) = g(q)^p$. The forms G_h, $E_{p-1}^m F$ and F^p all have weight $pm = m + h(p+1)$. By Lemma 3.1(i) and $E_{p-1} \equiv 1$, $D \equiv \theta^h F - \left(\frac{6}{p}\right)(F - \chi_{12}(p)F^p) \equiv 0$. □

Proof of Theorem 1.1. Let $E/\overline{\mathbb{F}}_p$ be supersingular and $\omega \neq 0$. Choose a level-12 structure α with Weil pairing ζ, and put $x = (E, \alpha, \omega)$. By Input S1 and Lemma 3.2, $D(x) = 0$. By Input S2, $E_{p-1}(x) = A(E,\omega) = 0$. Evaluating Lemma 3.1(ii) and the definition of D at x gives

$$\tilde{c}_h\,B(E,\omega)^h\,F(x) = -\left(\frac{6}{p}\right)\chi_{12}(p)\,F(x)^p.$$

By (3.1), $F(x)^{24} = \Delta(E,\omega)^{p+1} \neq 0$, so we may divide by $F(x)$. Using $F(x)^{p-1} = \Delta(E,\omega)^k$ and Lemma 3.1(iii),

$$B(E,\omega)^h = \left(\frac{6}{p}\right)\chi_{12}(p)\,12^h\,h!\,\Delta(E,\omega)^k.$$

Finally $12^h \equiv \left(\frac{12}{p}\right) = \left(\frac{3}{p}\right)$ by Euler's criterion, and $\chi_{12}(p) = \left(\frac{12}{p}\right) = \left(\frac{3}{p}\right)$ for $p \geq 5$. Hence $\chi_{12}(p)12^h = 1$, which proves (a). Part (b) follows from Lemma 2.2. □

Remark 3.3 (sources of the factorial and the sign). The factorial is the Pochhammer symbol $(m)_h \equiv (p-1)!/h!$, the leading coefficient of θ^h on weight $m \equiv \frac{1}{2} \pmod p$. The weight $(p+1)/2$ is *self-dual* for Serre's pairing $w \leftrightarrow p+1-w$. For even w this is Robert's Théorème 2 [8, §2.5, p. 119]: $(p-w)!\,G_w^p$ and $(E_{p+1}/12)^{w-1}G_{p+1-w}$ have the same image in $\mathcal{M}/(p, E_{p-1})$, a functional equation communicated by Serre. At level one a self-dual Eisenstein series exists only when $(p+1)/2$ is even. The form η^{p+1} supplies a self-dual form for every p, and its twist is explicit because $\eta^{p+1} \equiv \eta(\tau)\eta(p\tau)$ has "CM by $\mathbb{Q}(\sqrt{-p})$" modulo p. For $p \equiv 23 \pmod{24}$, $F = \Delta^{(p+1)/24}$ is of level one, and Lemma 3.2 contains Wilton's congruence $\tau(n) \equiv 0 \pmod{23}$ for $\left(\frac{n}{23}\right) = -1$. The symbol $\left(\frac{6}{p}\right)$, not $\left(\frac{-6}{p}\right)$, arises as $\left(\frac{24}{p}\right)$ from the exponent $1/24$ of η. The factor $\chi_{12}(p)12^h = 1$ removes all dependence on p mod 12.

Remark 3.4 (on the traps listed in the assignment). No evaluation of a q-expansion at a supersingular point occurs. The only q-expansion argument, Lemma 3.2, compares forms of equal weight pm. It keeps the factor E_{p-1}^m, and Input S1 turns it into an identity of sections. Only then is $A = 0$ used. The sign is obtained directly, not as a square root of Baker's congruence. Neither Baker's congruence nor the manuscript's signed-residue theorem is used.

4 Explicit forms

4.1 The supersingular polynomial

Write $p - 1 = 12n + 4\delta + 6\varepsilon$ with $\delta, \varepsilon \in \{0, 1\}$ and let $\mathrm{ss}_p(j) \in \mathbb{F}_p[j]$ be the monic supersingular polynomial.

Corollary 4.1 (signed form of Kaneko–Zagier (40))**.** *For every supersingular $j_0 \notin \{0, 1728\}$,*

$$\mathrm{ss}_p'(j_0)^{\frac{p-1}{2}}\, j_0^{\frac{(1-\delta)(p-1)}{3}}\, (j_0 - 1728)^{\frac{(1-\varepsilon)(p-1)}{4}} = \left(\frac{-2}{p}\right)\left(\frac{p-1}{2}\right)!.$$

Squaring gives [4, (40)] in the form proved by Baker [1, (6.1)]. In particular, for $p \equiv -1 \pmod{12}$ one gets $\mathrm{ss}_p'(j_0)^{(p-1)/2} = \pm 1$, hence $\mathrm{ss}_p'(j_0) \in \mathbb{F}_p$. This is the "simple statement" for which [4, §10] asked for an elementary proof.

Proof. $A = E_4^\delta E_6^\varepsilon \Delta^n s(j)$ with $\mathrm{ss}_p = j^\delta (j - 1728)^\varepsilon s$, and $B = 12\partial A$ [1, Prop. 1.3]; this also follows from Inputs S3 and S1 at level one. Since $\partial j = \theta j = -E_4^2 E_6/\Delta$, at a root j_0 of s we get $B = -12E_4^{\delta+2}E_6^{\varepsilon+1}\Delta^{n-1}s'(j_0)$. Raise this to the hth power and use $E_4^3 = j\Delta$ and $E_6^2 = (j - 1728)\Delta$; the exponents are integral because $3 \mid h(\delta+2)$ and $2 \mid h(\varepsilon+1)$. This gives $c = \left(\frac{-3}{p}\right) s'(j_0)^h j_0^{h(\delta+2)/3}(j_0 - 1728)^{h(\varepsilon+1)/2}$. Substitute $s'(j_0) = \mathrm{ss}_p'(j_0) j_0^{-\delta}(j_0 - 1728)^{-\varepsilon}$ and $c = \left(\frac{6}{p}\right) h!$, and note $\left(\frac{-3}{p}\right)\left(\frac{6}{p}\right) = \left(\frac{-2}{p}\right)$. □

4.2 Weierstrass models

For $f(x) = x^3 + ax + b$ put $c_i(a, b) = [x^i]\, f(x)^h$. On $(y^2 = f(x), dx/y)$ one has $E_4 = -3a$, $E_6 = 27b/2$ and $\Delta = -(4a^3 + 27b^2)/256$.

Proposition 4.2. *In $\mathbb{F}_p[a, b]$: $A = c_{p-1}$ and $B = 3\,c_{p-2}$. In particular, for every supersingular $y^2 = x^3 + ax + b$ over $\overline{\mathbb{F}}_p$,*

$$\left([x^{p-2}](x^3 + ax + b)^{\frac{p-1}{2}}\right)^{\frac{p-1}{2}} = \left(\frac{2}{p}\right)\left(\frac{p-1}{2}\right)!\left(-\frac{4a^3 + 27b^2}{256}\right)^{\frac{p^2-1}{24}}.$$

Proof. c_{p-1} is isobaric of weight $p - 1$ in (a, b) of weights $(4, 6)$, and it vanishes exactly at supersingular curves [10, Thm. V.4.1(a)]. By Lemma 2.2, $c_{p-1} = \kappa A$ with $\kappa \in \mathbb{F}_p$. At $(E_4, E_6) =$

$(1,1)$, i.e. $f=(x-\frac{1}{3})^2(x+\frac{2}{3})$, $c_{p-1}=\sum_s \binom{p-1}{s}(-\frac{1}{3})^s\binom{h}{s}(\frac{2}{3})^{h-s}\equiv(\frac{1}{3}+\frac{2}{3})^h=1$. Also $A(1,1)=1$, since that is the constant term of E_{p-1}. Hence $\kappa=1$.

Next, $B=12\partial A$, and $\partial a=\frac{3b}{2}$, $\partial b=-\frac{a^2}{3}$. Let $d_i=[x^i]f^{h-1}$. Then $12\partial c_{p-1}=h(18b\,d_{p-2}-4a^2d_{p-1})\equiv -9b\,d_{p-2}+2a^2d_{p-1}$. Compare coefficients of x^j in $f\cdot(f^{h-1})'=(h-1)f'f^{h-1}$, with $h-1\equiv-\frac{3}{2}$: $(2j+5)d_{j-2}+a(2j+3)d_j+2b(j+1)d_{j+1}\equiv 0$. For $j=p-1$ this gives $a\,d_{p-1}\equiv -3d_{p-3}$; for $j=p-3$ it gives $d_{p-5}+3a\,d_{p-3}+4b\,d_{p-2}\equiv 0$. Hence $12\partial c_{p-1}-3c_{p-2}=-6a\,d_{p-3}-12b\,d_{p-2}-3d_{p-5}-3a\,d_{p-3}=-3(d_{p-5}+3a\,d_{p-3}+4b\,d_{p-2})\equiv 0$, using $c_{p-2}=d_{p-5}+a\,d_{p-3}+b\,d_{p-2}$. The displayed formula is Theorem 1.1(a) with $\left(\frac{3}{p}\right)\left(\frac{6}{p}\right)=\left(\frac{2}{p}\right)$. □

Proposition 4.2 identifies B at supersingular points with $3c_{p-2}$. Up to a pth root, c_{p-2} is the coefficient of dx/y in the Cartier image of $x\,dx/y$. Translating x changes c_{p-2} by a multiple of c_{p-1}, so at supersingular points the formula holds for any monic cubic.

4.3 The Legendre family

Let $H(\lambda)=\sum_{i=0}^{h}\binom{h}{i}^2\lambda^i$ be the Deuring polynomial. On $\Gamma(2)$ one has $\theta\lambda=\frac{1}{2}\lambda(1-\lambda)\theta_3^4$ and $\Delta=2^{-8}\lambda^2(1-\lambda)^2\theta_3^{24}$, and $A=H(\lambda)\,\theta_3^{2(p-1)}$ modulo p. The last identity is the classical congruence $\theta_3^2={}_2F_1(\frac{1}{2},\frac{1}{2};1;\lambda)\equiv H(\lambda)\,\theta_3^{2p}$ together with Input S1. At a root λ_0 of H this gives $B=12\theta(H(\lambda))\theta_3^{2(p-1)}=6\lambda_0(1-\lambda_0)H'(\lambda_0)\theta_3^{2(p+1)}$, and Theorem 1.1 becomes:

Corollary 4.3. *For every root $\lambda_0\in\mathbb{F}_{p^2}$ of H,*

$$H'(\lambda_0)^{\frac{p-1}{2}}=\Big(\frac{p-1}{2}\Big)!\,2^{-\frac{p^2-1}{3}}\,\big(\lambda_0(1-\lambda_0)\big)^{\frac{(p-1)(p-5)}{12}}.$$

This corollary depends on the cited theta identities. It is also confirmed exactly for all $p\le 997$ (route II below). Combining Theorem 1.1 with Landweber's half-exponent identity [6, Theorem 3, p. 72] gives, for supersingular $y^2=4\prod_{i=1}^3(x-e_i)$ with $\omega=dx/y$, $E_{p+1}(E,\omega)^{(p-1)/2}=\left(\frac{6}{p}\right)h!\,(e_\alpha-e_\beta)^{(p^2-1)/4}$ for $\alpha\neq\beta$.

5 What is proved, imported and computed

Statement	Status
Theorem 1.1 (Conjecture 1.5), all $p\ge 5$	proved here, given Inputs S1–S5
Lemma 2.1 (η^2 on $\Gamma(12)$)	proved here, given the Petersson formula; checked numerically
Lemma 2.2 (equivalence, $j=0,1728$)	proved here, given squarefreeness of A (Input S2)
Constancy, Baker, eta character	corollaries; Robert's identity imported for (iii)
Corollary 4.1, Proposition 4.2	proved here; checked exactly
Corollary 4.3	imports classical $\Gamma(2)$ identities; checked exactly, $p\le 997$
q-expansion principle, Hasse invariant, Kummer, Serre derivative, Euler	imported (standard)
Conjectural content remaining	none for Conjecture 1.5

6 Computational verification

All computations use exact arithmetic in $\mathbb{F}_p$ and $\mathbb{F}_{p^2}$ (PARI/GP 2.17.2 via cypari2 2.2.4, Python 3.13). This implementation is independent of the inherited code: $E_{p\pm1}$ are found by solving the q-expansion system with PARI's `bernfrac` and `matsolvemod`. For each of the 166 primes $5\le p\le 997$ (36 with $p\equiv 1$, 44 with $p\equiv 5$, 44 with $p\equiv 7$, 42 with $p\equiv 11 \pmod{12}$), the following hold.

(IV) Divisibility: $R(j) \bmod \mathrm{ss}_p(j)$ is the constant $\left(\frac{6}{p}\right) h!$ (Lemma 2.2).

(I) Evaluation of the polynomial $E_{p+1}(E_4, E_6)$ at each of the 6430 supersingular j-values, 4734 of them not in $\mathbb{F}_p$, on three randomly rescaled models $(u^4 E_4, u^6 E_6)$, $u \in \mathbb{F}_{p^2}^\times$, each.

(II) The Legendre route of Corollary 4.3 at all roots of H, which uses no Eisenstein series. The resulting j-set agrees with the roots of ss_p.

(III) The Hasse–Witt route of Proposition 4.2, which uses no E_{p+1}; the identity $B = 3c_{p-2}$ is checked at all supersingular models.

A negative control replacing $\left(\frac{6}{p}\right) h!$ by $-\left(\frac{6}{p}\right) h!$ fails every sign-sensitive check. The supersingular counts agree with the Eichler–Deuring formula. For $p \in \{23, 47, 71, 167\}$, where F has level one, Lemmas 3.1 and 3.2 are checked as identities in $\mathbb{F}_p[E_4, E_6]$ (`mechanism_check.py`). For $p \le 200$ the polynomial identities $A = c_{p-1}$ and $B = 3c_{p-2}$ in $\mathbb{F}_p[a, b]$ are checked, and for $p \le 997$ Corollaries 4.1 and 4.3. The inherited script reproduces 166 matches through 997. Finite checks support the proof but are not part of it.

7 Literature and priority

Sources examined, all with explicit statements read:

- Baker [1]: Theorem 1.4 (the square), Proposition 1.3 ($B \equiv \partial A$), (6.1). There is no half-exponent statement and no factorial.
- Robert [8]: Théorème B (isogeny identity) and Théorème 2 (Serre's functional equation for G_w, $4 \le w \le p-3$ even). This is the same mechanism as here, but for Eisenstein series; no η and no evaluation of c_p.
- Landweber [6]: Theorems 1–3, the half exponent for Δ via $e_\alpha - e_\beta$. Nothing on E_{p+1}.
- Kaneko–Zagier [4], §10: (39)–(40) in squared form; asks for an elementary proof when $p \equiv -1 \pmod{12}$.
- Ono [7]: Theorem 1.2 and Remark 4 (squared isogeny ratios). No E_{p+1}^h.
- Recent papers found by search: Bringmann–Guerzhoy–Kane–Mertens–Rolen (2026), Ringeling (2022), and arXiv:2608.21198 (2026) on Legendre polynomials and CM. None contains the statement.

De Shalit's paper is known to us only through [4] and was not checked. No published signed evaluation of c_p was found in these sources. Because the argument combines standard tools in a short way, it may be known to specialists; priority should not be asserted without a wider search.

8 Suggested replacement text for the manuscript

(Offered for consideration; the inherited manuscript has not been edited.) Replace Conjecture 1.5 by:

> **Theorem 1.5.** $c_p = \left(\frac{6}{p}\right) \left(\frac{p-1}{2}\right)! \bmod p$; equivalently $E_{p+1}^{(p-1)/2} \equiv \left(\frac{6}{p}\right) \left(\frac{p-1}{2}\right)! \Delta^{(p^2-1)/24} \pmod{(p, E_{p-1})}$.

Replace the remark "on Conjecture 1.5" in Section 6 by a proof sketch:

Let $F = \eta^{p+1} \in M_{(p+1)/2}(\Gamma(12))$. Since $F \equiv \eta(\tau)\eta(p\tau) \pmod{p}$ and $\eta = \sum \chi_{12}(n) q^{n^2/24}$, the q-expansion principle gives $\theta^{(p-1)/2} F = \left(\frac{6}{p}\right)(A^{(p+1)/2} F - \chi_{12}(p) F^p)$ as mod-p forms of weight $p(p+1)/2$. On the other hand, iterating $\theta f = A\partial f + \frac{w}{12} B f$ gives $\theta^{(p-1)/2} F \equiv -\frac{1}{h!}(B/12)^h F \pmod{A}$. At a supersingular point $A = 0$ and $F \neq 0$, so $(B/12)^h = \left(\frac{6}{p}\right) \chi_{12}(p) h!\, F^{p-1} = \left(\frac{6}{p}\right) \chi_{12}(p) h!\, \Delta^k$, and $\chi_{12}(p) 12^h = 1$.

The proof gives Corollary 1.4 and Theorem 1.3 independently, so the authors may also note that these results, and with them Theorem 1.1(e), no longer depend on Baker's congruence.

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